

AGILE SPRINT PLANNING

Title - AI RESUME ANALYSER

Group 30

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1. PROJECT OVERVIEW

Project Title: AI Resume Analyzer

The AI Resume Analyzer is an AI-powered web application designed to help job seekers and students improve their resumes and prepare for job opportunities. The system analyzes uploaded resumes, provides an ATS compatibility score and improvement suggestions, identifies missing skills and keywords for a target job, and generates personalized interview questions.

The project is developed using a modern web application architecture with **ASP.NET Core Web API, Entity Framework Core, React, SQL Server, and Google Gemini AI**. The development is divided into short Agile Sprints to implement, test, and integrate the system incrementally.

2. Agile Methodology

The **Agile methodology** is used to develop the AI Resume Analyzer through short and focused Sprints. Each Sprint addresses a specific set of features and technical tasks from the Product Backlog.

The project follows the cycle:

Planning → Development → Testing → Review → Integration

This approach allows the team to develop the project incrementally, identify issues early, integrate completed features, and ensure that the final system is completed within the planned **10-day development timeline**.

3. Product Vision

Product Vision Statement:

“To build an intelligent and user-friendly Resume Analyzer that helps job seekers create stronger, ATS-friendly resumes, identify job-specific skill gaps, and prepare effectively for interviews through AI-powered analysis and personalized recommendations.”

The product aims to provide users with a single platform for **resume analysis, job matching, improvement recommendations, and personalized interview preparation**.

4. Product Backlog

The Product Backlog contains the prioritized features and technical tasks required to develop the AI Resume Analyzer. The backlog is organized according to the development flow and distributed across the planned Sprints.

Backlog ID	Backlog Item	Priority	Planned Sprint
PB-01	Set up ASP.NET Core Web API project	High	Sprint 1
PB-02	Configure Entity Framework Core and SQL Server	High	Sprint 1
PB-03	Create User and Resume entities	High	Sprint 1
PB-04	Integrate Google Gemini AI service	High	Sprint 1
PB-05	Implement user registration and secure login	High	Sprint 2
PB-06	Implement resume upload and file validation	High	Sprint 2
PB-07	Implement resume text extraction	High	Sprint 2
PB-08	Implement ATS score and resume analysis	High	Sprint 3
PB-09	Generate resume improvement recommendations	High	Sprint 3
PB-10	Implement target job description input	High	Sprint 4
PB-11	Implement job matching and skill gap analysis	High	Sprint 4
PB-12	Identify missing skills and keywords	High	Sprint 4
PB-13	Generate Technical, Project, and HR interview questions	High	Sprint 5
PB-14	Implement analysis history and report management	Medium	Sprint 6
PB-15	Implement PDF report generation/download	Medium	Sprint 6
PB-16	Integrate React frontend with backend APIs	High	Sprint 6
PB-17	Perform system testing, bug fixing, and final integration	High	Sprint 6
PB-18	Deploy and prepare the final application	High	Sprint 6

5. Sprint Planning

Sprint Planning defines how the Product Backlog is divided into manageable development Sprints. The AI Resume Analyzer is planned for completion within an approximately **10-day development timeline**, with each Sprint focusing on a specific technical and functional area.

Sprint Schedule

Sprint	Timeline	Sprint Focus	Main Backlog Items
Sprint 1	Day 1–2	Backend, Database & AI Foundation	PB-01 to PB-04
Sprint 2	Day 3–4	Authentication & Resume Upload	PB-05 to PB-07
Sprint 3	Day 5–6	ATS Analysis & Recommendations	PB-08 to PB-09
Sprint 4	Day 7	Job Matching & Skill Gap Analysis	PB-10 to PB-12
Sprint 5	Day 8	Interview Question Generation	PB-13
Sprint 6	Day 9–10	Frontend Integration, Reports, Testing & Deployment	PB-14 to PB-18

Sprint Planning Process

For each Sprint, the team will:

1. Select the relevant items from the Product Backlog.
2. Define the Sprint Goal and User Stories.
3. Break the stories into development tasks.
4. Implement and test the assigned features.
5. Review the completed work.
6. Integrate the completed features with the existing system.

Each Sprint builds on the previous Sprint so that the complete application can be developed, tested, and prepared for final delivery within the project timeline.

6. Sprint Details

This section describes the work planned for each Sprint of the **AI Resume Analyzer** project. Each Sprint focuses on specific features from the Product Backlog and includes a defined goal, tasks, expected outcome, and deliverable.

The six Sprints are planned according to the project's **10-day development timeline**, with each Sprint building upon the functionality completed in the previous one.

The Sprints cover:

- **Sprint 1:** Project Setup & User Authentication
- **Sprint 2:** Resume Upload & Text Extraction
- **Sprint 3:** ATS Analysis & Recommendations
- **Sprint 4:** Job Matching & Skill Gap Analysis
- **Sprint 5:** Interview Question Generation
- **Sprint 6:** Reports, Testing & Final Integration

Each Sprint follows the cycle of **Planning→Development→Testing→Review→Integration**.

Sprint 1 – Backend, Database & AI Foundation

Timeline: Day 1–2

Sprint Goal: Initialize the ASP.NET Core Web API, configure Entity Framework Core with SQL Server, create the core entities, and establish the Gemini AI integration.

User Stories

Story 1: As a backend developer, I need to create the ASP.NET Core Web API so that the frontend can communicate with the backend services.

Story 2: As a database developer, I need to configure Entity Framework Core and define the required entities so that application data can be stored and managed.

Story 3: As an AI developer, I need to integrate the Gemini AI service so that resume content can be analyzed using AI.

Development Tasks

Task ID	Development Task
T1	Create and configure the ASP.NET Core Web API project
T2	Configure Entity Framework Core
T3	Configure SQL Server database connection
T4	Create User and Resume entities
T5	Create and apply database migrations
T6	Configure Gemini AI integration
T7	Test API, database, and AI connectivity

Expected Outcome

A functional backend foundation with **ASP.NET Core Web API, SQL Server database connectivity, core entities, and Gemini AI integration**.

Sprint Deliverable

A working backend and AI foundation ready for implementing authentication and resume-processing features in Sprint 2.

Sprint 2 – Authentication & Resume Upload

Timeline: Day 3–4

Sprint Goal: Implement secure user authentication and enable users to upload valid resume files for further processing and analysis.

User Stories

Story 1: As a user, I want to register and log in securely so that I can access the application and my resume analysis.

Story 2: As a user, I want to upload my resume so that the system can process it for analysis.

Story 3: As a user, I want invalid or unsupported files to be rejected so that only valid resumes are processed.

Development Tasks

Task ID	Development Task
T1	Implement user registration API
T2	Implement secure login and authentication
T3	Create the resume upload interface
T4	Implement resume upload API
T5	Add PDF/DOCX file validation
T6	Handle unsupported and invalid files
T7	Test authentication and resume upload functionality

Expected Outcome

Users can securely register and log in, upload a valid resume, and receive appropriate validation messages for unsupported or invalid files.

Sprint Deliverable

A working authentication module and resume upload functionality ready for resume parsing and ATS analysis in Sprint 3.

Sprint 3 – ATS Analysis & Recommendations

Timeline: Day 5–6

Sprint Goal: Implement resume parsing and AI-powered ATS analysis to evaluate the resume and provide meaningful improvement recommendations.

User Stories

Story 1: As a user, I want the system to extract my resume content so that it can be analyzed automatically.

Story 2: As a user, I want to receive an ATS compatibility score so that I can understand how well my resume performs against ATS requirements.

Story 3: As a user, I want to receive strengths, gaps, and improvement recommendations so that I can improve my resume.

Development Tasks

Task ID	Development Task
T1	Implement PDF/DOCX text extraction
T2	Process and structure extracted resume content
T3	Connect resume data with the Gemini AI service
T4	Implement ATS scoring logic
T5	Analyze resume sections, skills, formatting, and content
T6	Generate strengths and identified gaps
T7	Generate actionable improvement recommendations
T8	Test ATS analysis with sample resumes

Expected Outcome

The system can extract resume content, generate an ATS score, identify strengths and gaps, and provide actionable recommendations.

Sprint Deliverable

A functional **ATS analysis module** that provides users with a resume score, analysis, and improvement recommendations.

Sprint 4 – Job Matching & Skill Gap Analysis

Timeline: Day 7

Sprint Goal: Implement job-specific resume analysis to compare the user's resume with a target job description and identify relevant and missing skills and keywords.

User Stories

Story 1: As a user, I want to enter a target job description so that my resume can be evaluated against a specific job.

Story 2: As a user, I want to see the skills and keywords that match the job description so that I can understand my suitability for the role.

Story 3: As a user, I want to identify missing skills and keywords so that I can improve my resume according to the target job.

Development Tasks

Task ID	Development Task
T1	Create job description input functionality
T2	Validate the submitted job description
T3	Process the job description using the AI service
T4	Compare resume content with job requirements
T5	Identify matching skills and keywords
T6	Identify missing skills and keywords
T7	Display job-specific matching results
T8	Test job matching with different job descriptions

Expected Outcome

The system can compare a resume with a target job description and clearly display matching and missing skills and keywords.

Sprint Deliverable

A functional **Job Matching and Skill Gap Analysis module** that helps users understand how well their resume matches a specific job role.

Sprint 5 – Interview Question Generation

Timeline: Day 8

Sprint Goal: Implement AI-powered interview preparation by generating personalized Technical, Project, and HR questions based on the user's resume.

User Stories

Story 1: As a user, I want to generate interview questions from my resume so that I can prepare for interviews based on my own experience.

Story 2: As a user, I want to receive Technical and Project-related questions so that I can prepare for questions related to my skills and projects.

Story 3: As a user, I want to receive HR questions so that I can prepare for common behavioral interview discussions.

Development Tasks

Task ID	Development Task
T1	Create interview preparation functionality
T2	Send analyzed resume content to the Gemini AI service
T3	Generate Technical interview questions
T4	Generate Project-related interview questions
T5	Generate HR interview questions
T6	Structure and display generated questions
T7	Test questions using different resume profiles

Expected Outcome

The system generates relevant and personalized **Technical, Project, and HR interview questions** based on the user's resume.

Sprint Deliverable

A functional **AI Interview Preparation module** that provides personalized questions to help users prepare for interviews.

Sprint 6 – Frontend Integration, Reports, Testing & Deployment

Timeline: Day 9–10

Sprint Goal: Complete the React frontend integration, implement report management, integrate all developed modules, perform final testing, and prepare the application for deployment.

User Stories

Story 1: As a user, I want to access all resume analysis features through a simple interface so that I can easily use the application.

Story 2: As a user, I want to view my previous analysis reports so that I can track my resume evaluations.

Story 3: As a user, I want to download my analysis report so that I can save it for future reference.

Development Tasks

Task ID	Development Task
T1	Integrate React frontend with ASP.NET Core APIs
T2	Connect authentication and resume-analysis interfaces
T3	Integrate ATS, job matching, and interview modules
T4	Implement analysis history
T5	Implement PDF report generation and download
T6	Implement report deletion functionality
T7	Perform functional and integration testing
T8	Fix identified bugs and validation issues
T9	Perform final system testing
T10	Prepare and deploy the final application

Expected Outcome

All major modules are integrated into a single working application, with reports, testing, error handling, and final deployment completed.

Sprint Deliverable

A complete and tested **AI Resume Analyzer** ready for final demonstration and delivery by the project deadline.

7. Definition of Done

A Product Backlog item or Sprint task is considered **Done** when it has been fully implemented, tested, and integrated into the application.

Criteria	Definition
Development	The assigned functionality is completely implemented according to the requirement.
Testing	The functionality has been tested successfully with valid and invalid inputs.
Validation	Appropriate input validation and error handling are implemented.
Integration	The completed feature works correctly with the existing modules.
Code Quality	The code is organized, readable, and free from known critical errors.
UI Integration	The feature is accessible and functional through the application interface where applicable.
Review	The completed work has been reviewed and meets the Sprint requirements.
Documentation	Required technical or functional documentation is updated.
Final Acceptance	The feature works as expected and is ready for the next Sprint or final release.

8. Sprint Review

At the end of each Sprint, the completed work is reviewed to ensure that the planned goals and requirements have been achieved. The team demonstrates the implemented features, verifies their functionality, and identifies any issues that need to be addressed.

Sprint	Review Focus
Sprint 1	Verify API setup, database connectivity, core entities, and Gemini AI integration.
Sprint 2	Verify user registration, login, authentication, and resume upload functionality.
Sprint 3	Verify resume parsing, ATS score, analysis results, and improvement recommendations.

Sprint 4	Verify job description matching, skill matching, and missing keyword/skill identification.
Sprint 5	Verify generation of Technical, Project, and HR interview questions.
Sprint 6	Verify complete system integration, reports, frontend functionality, testing, and deployment.

Sprint Review Outcome: The team confirms that the completed features meet the Sprint objectives and are ready for integration into the next stage or final release.

9. Sprint Retrospective

At the end of each Sprint, the team reviews the development process to identify what went well, the challenges faced, and improvements needed for the next Sprint.

Sprint	What Went Well	Challenges / Improvements
Sprint 1	Backend, database, and AI foundation were established.	Resolve initial configuration and integration issues early.
Sprint 2	Authentication and resume upload functionality were implemented.	Improve file validation and error handling.
Sprint 3	Resume parsing and ATS analysis were integrated successfully.	Improve AI response accuracy and consistency.
Sprint 4	Job matching and skill gap analysis were implemented.	Improve matching accuracy for different job descriptions.
Sprint 5	Personalized interview questions were generated successfully.	Improve relevance and categorization of generated questions.
Sprint 6	Features were integrated, tested, and prepared for final delivery.	Fix remaining bugs and improve overall system stability.

Overall Retrospective

The team will use the feedback and issues identified during each Sprint to improve the development process, maintain code quality, and ensure timely completion of the project.

10. Final Project Outcome

The AI Resume Analyzer will provide an integrated platform that helps users analyze and improve their resumes and prepare for job opportunities.

At the end of the project, the system will provide:

- Secure user registration and login.
- Resume upload and text extraction.
- AI-powered ATS score and resume analysis.
- Resume strengths, gaps, and improvement recommendations.
- Job description matching with skill and keyword gap analysis.
- Personalized Technical, Project, and HR interview questions.
- Analysis history and downloadable PDF reports.
- A responsive React frontend integrated with the ASP.NET Core backend.
- Tested and integrated application ready for final deployment and demonstration.